

This is due by July 25, but I strongly encourage you to finish it before midterm 1.

Problem 1. Consider $\mathbb{Z}[\frac{1+\sqrt{-3}}{2}] = \{a+b\frac{1+\sqrt{-3}}{2} : a, b \in \mathbb{Z}\}$. This is a slightly larger set than $\mathbb{Z}[\sqrt{-3}]$.

1. Note that $w = \frac{1+\sqrt{-3}}{2}$ is a root of the equation $x^2 - x + 1$. Show $\mathbb{Z}[\frac{1+\sqrt{-3}}{2}]$ is a ring.
2. Define $|a + b\frac{1+\sqrt{-3}}{2}| = a^2 + ab + b^2$. Show that $|-|$ is a norm function $\mathbb{Z}[\frac{1+\sqrt{-3}}{2}] \rightarrow \mathbb{N}$.
3. Show $|xy| = |x||y|$ for all $x, y \in \mathbb{Z}[\frac{1+\sqrt{-3}}{2}]$. That is, it is a multiplicative norm.
4. Find all the units in $\mathbb{Z}[\frac{1+\sqrt{-3}}{2}]$.
5. Determine and explain if $\mathbb{Z}[\frac{1+\sqrt{-3}}{2}]$ is a domain.

Proof. 1. Use $w^2 - w + 1$ or equivalently $w^2 = w - 1$ to eliminate higher powers of w . This would help you to simplify the addition and multiplication.

2. One need to show that the norm is nonnegative. (Fred.) Following the steps of completing the squares, $a^2 + ab + b^2 = a^2 + ab + \frac{1}{4}b^2 + \frac{3}{4}b^2 = (a + \frac{1}{2}b)^2 + \frac{3}{4}b^2 \geq 0$ for all $a, b \in \mathbb{Z}$.
3. In general, if w is a root to the equation $x^2 - \alpha x + \beta$ for $\alpha, \beta \in \mathbb{Z}$, then $\mathbb{Z}[w] = \{a + bw : a, b \in \mathbb{Z}\}$ is a ring, which is often called a quadratic integer ring. We can define $|a + bw| = (a + bw)(a + b\bar{w})$ where $\bar{w}$ is the complex conjugate of w . Think about why $|-|$ is a multiplicative function $\mathbb{Z}[w] \rightarrow \mathbb{Z}$. What does that tell you about the norm of the product of two complex numbers?
4. Note that $a + bw$ is a unit implies $|a + bw| = a^2 + ab + b^2 = 1$. But this is only a necessary (and not-known-to-be-sufficient yet) condition. But $a^2 + ab + b^2 = 1$ only has so few integer solutions such units must lie in $\{\pm 1, \pm w, \pm(w - 1)\}$ (also one may write $\pm(w - 1) = \pm w^2$). Clearly ± 1 are units. Note that $w^2 - w + 1 = 0$ implies $1 = w(1 - w) = -w(w - 1)$ which implies both $\pm w$ and $\pm w - 1$ are units.
5. (Marcial, Tianshuo, Albert.) Note the norm is nonnegative and note that $|a + bw| = 0$ iff $a = b = 0$ iff $a + bw = 0$. So for any $x, y \in \mathbb{Z}[\frac{1+\sqrt{-3}}{2}]$ $xy = 0$ iff $N(x)N(y) = 0$ iff $N(x) = 0$ or $N(y) = 0$ (since $\mathbb{Z}$ is a domain) iff $x = 0$ or $y = 0$.

(Jenny.) Note that $\mathbb{Z}[\frac{1+\sqrt{-3}}{2}] \subseteq \mathbb{Q}[\sqrt{-3}] = \mathbb{Q}(\sqrt{-3})$ which is a subring of a field.

(Nikoli.) Note that $\mathbb{Z}[\frac{1+\sqrt{-3}}{2}] \subseteq \mathbb{C}$ which is a subring of a field. □

Problem 2. Let R be a commutative ring.

1. Let $0 \neq r \in R$. Show r is irreducible iff (r) is maximal among the principal ideals.
2. Suppose $R = \mathbb{Z}_6$. Show $[2]$ is prime.
3. Suppose $R = \mathbb{Z}_6$. Show $[2]$ is not irreducible.

- Proof.* 1. Note that r is irreducible iff $r = ab$ implies a or b is a unit iff $(r) \subseteq (a)$ implies $(a) = (r)$ (if b is a unit) or $(a) = R$ if a is a unit. This is equivalent to say r is irreducible iff (r) is maximal among the principle ideals.
2. This question is simple if we follow the definitions step by step. (Nikoli.) Suppose $[2] \mid [a][b]$. Then $2 \equiv ab \pmod{6}$. That is $6 \mid ab - 2$. It implies $2 \mid ab - 2$ implies $2 \mid ab$ which shows $2 \mid a$ or $2 \mid b$. Clearly $[2] \mid [a]$ or $[2] \mid [b]$.
3. Note that $[2] = [2][4]$ where both $[2]$ and $[4]$ is not a unit nor zero. $\square$

Problem 3. Consider $R = \mathbb{Z}[\sqrt{-5}]$. You may assume it is a domain.

1. Write down a multiplicative norm.
2. Classify all the units in R .
3. Is 3 an irreducible element in R ?
4. Factorize 9 in two ways and determine if R is an UFD.
5. Is 3 a prime element in R ?

- Proof.* 1. The norm is from Problem 1. That is, $|a + b\sqrt{-5}| = a^2 + 5b^2$.
2. The units again must be of norm 1 which are necessarily ± 1 , which are clearly units.
3. Suppose $3 = xy$ into nonunits. Then necessarily $|x| = |y| = 3$ otherwise one of them has norm 1 and hence must be a unit. Write $x = a + b\sqrt{-5}$. Then $|x| = a^2 + 5b^2 = 3$. Mod 5, we get $a^2 \equiv 3 \pmod{5}$. But $a^2 \equiv 1$ or $4 \pmod{5}$ in general.
4. Note $9 = 3 \times 3 = (2 + \sqrt{-5})(2 - \sqrt{-5})$. We have seen 3 is irreducible. We show $2 + \sqrt{-5}$ is irreducible. But the norm is still 3 and by the same argument, $2 + \sqrt{-5}$ is also irreducible. Then 9 can be factored into irreducibles differently and $\mathbb{Z}[\sqrt{-5}]$ is not an UFD.
5. We note 3 is not a prime because $3 \mid (2 + \sqrt{-5})(2 - \sqrt{-5})$ and yet $3 \nmid 2 \pm \sqrt{-5}$. Indeed, if $3(a + b\sqrt{-5}) = 2 \pm \sqrt{-5}$, then $3a = 2$ in $\mathbb{Z}$ which is impossible. $\square$